SIGNAL SCHOOL
ITG, ETC, PA

SIGNAL ESTIMATE



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NEED

Signal Estimates is crucial for determining the required communication resources, ensuring signal coverage, identifying potential interference, and supporting effective planning and execution of military operations.

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55

TIME

16 HOURS

ROADMAP

DIAGNOSE	EXPERIMENT	REFLECT	EVALUATION
• Signal Estimate and importance • Signal Estimate Steps	Check on learning through asking questions	Students to summarize the lesson	Group Presentation

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LEARNING OUTCOME

At the end of this subject, the students will be able to determine the steps of signal estimate in signal planning in accordance with PAM 6-01202-3. (C3)

REFERENCES

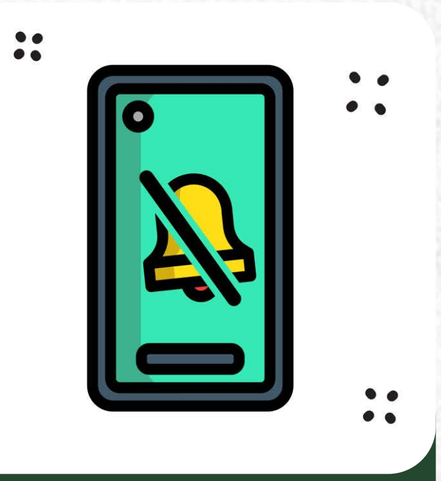
- PAM 6-0102 – Signal Operations



CLASSROOM RULES



QUESTION?
RAISE YOUR HAND



PHONE ON
SILENT NODE



COFFEE BREAK



NO SLEEPING



STRETCHING

SIGNAL ESTIMATE

- Signal estimate is an orderly and logical examination of all factors, known or assumed, that may affect the successful accomplishment of signal support operations mission.

SIGNAL ESTIMATE

- Signal officers are responsible to have a clear understanding of the type of environment the Army units will be operating in and its effect on various signal equipment.
- The Signal Estimate sets the stage for signal support operations.
- It allows signal operators to determine the appropriate signal support strategy that is going to be applied to the operations.
- Signal estimate is as dynamic as the combat operations itself and may change from time to time.

SIGNAL ESTIMATE

- The keyword “METAL” aids the Signal planner in the preparation of the signal estimate in any given situation and location. This keyword takes into account the “Mission” in precise terms, the “Enemy” signal capabilities, the “Troops” available to accomplish the mission, the “Area of Operations”, the “Logistical” requirements to assure the successful accomplishment of the mission. Signal estimate is also referred to as C4S Estimate.

STEPS IN MAKING ESTIMATE

- Know the Problem
- Gather Data and determine possible Solutions
- Analyze the Solutions
- Compare the Solutions
- Decide / Recommend

STEPS IN MAKING ESTIMATE

- MISSION
- SITUATION AND COURSES OF ACTION
- ANALYSIS OF OPPOSING COURSES OF ACTION
- COMPARISON OF OWN COURSES OF ACTION
- DECISION / RECOMMENDATION

MISSION

Signal mission must base from the mission of the unit concept of operation.

SITUATION AND COURSES OF ACTION

- Consider known & assumed factors & aspects that will affect the operation.
- contains 4 subparagraphs; Characteristics of the area of operation, Enemy situation, Own situation and Courses of Action

ANALYSIS OF OPPOSING COURSES OF ACTION

Evaluation of each COA that may have significant influence or have little or no effect upon the success of the operation.

COMPARISON OF OWN COURSES OF ACTION

Comparison of COA that will be deployed to provide commanders with adequate command and control.

DECISION / RECOMMENDATION

Selection of COA that has the greatest possibility assuring the success of the mission.

SUMMARY

- Mission
- Situation and Courses of Action
- Analysis of Opposing Courses of Action
- Comparison of Own Courses of Action
- Decision / Recommendation

SIGNAL ESTIMATE FORMAT

Para I - MISSION

Para II - SITUATION AND COURSES OF ACTION

Para III - ANALYSIS OFOPPOSING COURSES OF ACTION

Para IV - COMPARISON OF OWN COURSES OF ACTION

Para V - DECISION / RECOMMENDATION

Annex A
Signal Estimate Format

I MISSION
II SITUATION AND COA
AO and AI Analysis
A. TERRAIN:
B. WEATHER:
C. EXISTING COMMUNICATIONS:
D. ENEMY:
a. Organizational Structure:
b. Disposition and Location:
c. Strengths and Weaknesses:
d. Capabilities and Limitation:
E. OWN SITUATION:
a. Organizational Structure of Forces:
b. Disposition and Location of Friendly Forces:
c. Status of Personnel and Training Profile (ASSUMED):
d. Inventory of Signal Assets (ASSUMED):
e. Capabilities:
F. COURSES OF ACTION:
III ANALYSIS OF OPPOSING COA
A. COMMUNICATION AND DISPOSITION OF OPPOSING FORCES:
B. SIGNAL PROPAGATION OF OPPOSING FORCES:
C. LIST OF ENEMY CAPABILITIES AND COUNTERMEASURE (RFI from Intel):
Capabilities:
Limitations:
D. LIST OF FRIENDLY FORCES CABABILITIES AND LIMITATION:
Capabilities:
Limitations:
E. COUNTER-MEASURES:
On the Enemy Capabilities:
Signal Platoon Limitations:
IV COMPARISON OF COAs
V DECISION/RECOMMENDATIONS:
A. PERSONNEL REQUIREMENTS:
RECOMMENDATIONS:
B. C4S REQUIREMENTS:
RECOMMENDATIONS:
C. SUSTAINMENT REQUIREMENTS:
RECOMMENDATION:
D. MEANS OF COMMUNICATION:
E. LOCATION OF TCP AND MCP:
F. TASK LIST (Implied and Specific) (CMDR'S DECISION)

PARA I - MISSION

To provide secured, reliable, responsive and uninterrupted communication support and LIMITED electronic warfare (support) to 201 Brigade, 2ID as SE in the conduct of IPSO to contain Mindoro Island from CY 2025 to 2026 in order to prevent the CTs from augmenting their forces in CALABARZON from 01 Jan 2025 to 2025.

PARA II - SITUATION AND COURSES OF ACTION

AO and AI Analysis
A. TERRAIN:
B. WEATHER:
C. EXISTING COMMUNICATIONS:
D. ENEMY:
    a. Organizational Structure:
    b. Disposition and Location:
    c. Strengths and Weaknesses:
    d. Capabilities and Limitation:
E. OWN SITUATION:
    a. Organizational Structure of Forces:
    b. Disposition and Location of Friendly Forces:
    c. Status of Personnel and Training Profile (ASSUMED):
    d. Inventory of Signal Assets (ASSUMED):
    e. Capabilities:
F. COURSES OF ACTION:

PARA II - SITUATION AND COURSES OF ACTION

TERRAIN

The area of operations (AOR) is the entire island of Mindoro, which is separated from the Southern Luzon Mainland, it is composed of Occidental Mindoro and Oriental Mindoro. These two provinces are separated by a mountain range, running through the entire length of the island which serves as natural and political boundary.



PARA II - SITUATION AND COURSES OF ACTION

TERRAIN

The topography of the two provinces is generally rugged, with narrow strips of costal lowland. Its terrain is characterized by successive mountain range, valley and elongated plateaus, with rolling lands along the coastal region

This low lying alluvial plain will increase the range of our communication equipment. While the Area of operation has two (2) consider dead spots for the operating ground troops such as Mount Iglit in San Jose, Occidental Mindoro, Mount Halcon in Oriental Mindoro; other mountain ranges will pose a problem for both enemy and friendly communication systems for it will decrease the range of our VHF tactical radios and create dead spots for VHF tactical radios within the AO.



PARA II - SITUATION AND COURSES OF ACTION

TERRAIN

However, Mount Iglit, Occidental Mindoro, and Mount Halcon, Oriental Mindoro will serve as key terrains, will beep-up the range of our communication system provided that mobile/fixed relay/repeater stations will be established at these locations. Also, HF tactical radios will be of most important during operation underneath these mountains.



PARA II - SITUATION AND COURSES OF ACTION

WEATHER

The province is dry from November to April and wet during the rest of the year. Its coolest month are December to February; its warmest temperature is registered in March to June. Its average maximum temperature goes to 30ºC, its minimum humidity ranges from 63% to 87% with an annual average of 75.4%.

Cloud ceiling
Precipitation
Wind direction/speed
Humidity
Temperature
Visibility: Day Night

PARA II - SITUATION AND COURSES OF ACTION

WEATHER

- Thunderstorms and rain showers in early evening will limit the operations of the VHF tactical radios

- High temperature during day time will affect the Ionosphere consequently as to the modulating signal of HF radio.

PARA II - SITUATION AND COURSES OF ACTION

EXISTING COMMUNICATIONS

- The AO has one (1) existing Division Secured Mode VHF Repeater System (SMVRS) located at Pinamalayan Orr Min. while Telecommunications Companies have established cellular sites in various areas of the Island of Mindoro namely: PLDT-SMART and Globe Telecoms.
- Globe - GC 51P7K1234567890
- Smart - GC 51P7K1234567890

- UHF Repeater System at Mt. Halcon GC-_____
  VHF Repeater System at TCP GC - _____
  Radio room at TOC GC- _____

PARA II - SITUATION AND COURSES OF ACTION

ENEMY

A. ORGANIZATIONAL STRUCTURE

The enemy opposing the Brigade has been in the following positions for nearly 2 days. The 10th Light Combined Arms Brigade (CAB), 1st AG, MC1 Brigade Hqs is believed to be consolidated at vic Ft Magsaysay, Nueva Ecija (GS 9405). The 55th Mechanized Infantry Battalion, 10th Light CAB, 1st AG, MC1 is monitored at vicinity Sitio Labney (51P TT 302 156), San Jose, Tarlac, utilizing the CAB's engineer support in making road nets for its Mechanized assets' moving towards (GS 2613), Brgy Mababanaba, San Jose, Tarlac. It is expected to be in the vicinity of Brgy Mababanaba after four (4) days of continuous movement to establish a command post and subsequently cross designated LD/LC before its attack towards Brgy Iba to prevent reinforcements from CERAB.

The 8th Mechanized Infantry Battalion, 10th Light CAB, 1st AG, MC1 is monitored at vicinity 51P TT 336127, Brgy San Juan de Valdez, San Jose, Tarlac, believed to seize vicinity Brgy. Lubigan before advancing towards the west and supporting the subsequent seizure of CERAB. The 92nd Mechanized Infantry Battalion, 10th Light CAB, 1st AG, MC1 is monitored at vicinity 51PTT3356712701, Brgy San Juan de Valdez, Care, Tarlac along with 8th Mechanized Infantry Battalion, 10th Light CAB, 1st AG, MC1 and believed to utilize the Capas-Botolan road nets to advance towards CERAB The 12th Light CAB, 1st Army Group, MC1, NRA is currently in Sta. Fe, Nueva Vizcaya preparing its three (3) Mechanized Infantry Battalions to advance towards Alaminos, Pangasinan.

PARA II - SITUATION AND COURSES OF ACTION

ENEMY

B. DISPOSITION AND LOCATION

The 12th Light CAB, 1st Army Group, MC1, NRA in Sta. Fe, Nueva Vizcaya, is the TF3 confronted enemy unit. Its three (3) Mechanized Infantry Battalions, 55th Mechanized Infantry Battalion, 8th Mechanized Infantry Battalion, and 92nd Mechanized Infantry Battalion are preparing to advance towards Alaminos, Pangasinan.

PARA II - SITUATION AND COURSES OF ACTION

ENEMY

C. STRENGTHS AND WEAKNESSES

Based on latest monitoring, the enemy personnel are believed to be at 70% filled-up. Likewise, it is believed that 10th Light CAB is currently 40% filled up in terms of Armored Vehicles and artillery assets

PARA II - SITUATION AND COURSES OF ACTION

ENEMY

D. CAPABILITIES AND LIMITATION

The enemy is capable of battlefield Identification Friend or Foe (IFF) system and Satellite Navigation System, but limited electronic warfare support capabilities (Radio Direction Finder, Radio and Cellular Phone Communications Monitoring System), and defensive cyber warfare capabilities (Firewall and Secured Intranet). It has a 70% radio and wireless communication system.

PARA II - SITUATION AND COURSES OF ACTION

OWN SITUATION

A. ORGANIZATIONAL STRUCTURE OF FORCES

(Bde) One (1) BSSP
(Bn) One (1) Signal Pltn.

PARA II - SITUATION AND COURSES OF ACTION

OWN SITUATION

B. DISPOSITION AND LOCATION OF FRIENDLY FORCES

- Currently positioned at HTF3, Camp O'Donnel, Capas, Tarlac.
- A COY ----
- B COY -----
- C COY ----- (ALL UNITS)

PARA II - SITUATION AND COURSES OF ACTION

OWN SITUATION

C. STATUS OF PERSONNEL AND TRAINING PROFILE (ASSUMED)

Unit	Career Courses	Cyber Sec Trngs	TROMC	OPEC	IPEC	ISC	APSIOMT	CCNA	CSSC
HTF3	183	38	1	0	0	13	0	1	11
TOTAL	183	38	1	0	0	13	0	1	11

PARA II - SITUATION AND COURSES OF ACTION

OWN SITUATION

D. INVENTORY OF SIGNAL ASSETS (ASSUMED)

Nomenclature	Organic Units	OPCON Units	Total	Remarks
Communications	104	-		R1 – Highly Equipped
Sensor Systems	343	-		R1 – Highly Equipped
Common Operational Picture/Situational Awareness	16	-		R1 – Highly Equipped
Other C2 Systems	129	-		R1 – Highly Equipped

Nomenclature	Organic Units	OPCON Units	Total	Remarks
Communications				
Combat Net Radio				
HF 100-150W Base Radio	24	-		R1 – Highly Equipped
HF 20W Manpack Radio	22	-		R1 – Highly Equipped
VHF 2-10W Handheld Radio	51	-		R1 – Highly Equipped
UHF/FM 2-10W Handheld Radio	6	-		R1 – Highly Equipped
VHF Repeater System	1	-		R1 – Highly Equipped

PARA II - SITUATION AND COURSES OF ACTION

OWN SITUATION

E. CAPABILITIES

Strength and Capabilities:
a. Capable of installing Secured Mode VHF Repeater System (SMVRS).
b. Capable of deploying Mobile Communications Van.
c. Capable of conducting limited Electronic Warfare Protection to counter Electronic Warfare Attack of the EN in coordination with Intelligence Unit.
d. Capable in programming of radio frequency of Combat Net Radios using Communications Planning Application (CPA).
e. Capable of using Battlefield Management System to monitor real-time movement of operating troops
f. Capable to communicate in High Frequency (Harris Manpack), Very High Frequency (Harris Handheld), Ultra High Frequency (Commercial Radios) spectrum.

PARA II - SITUATION AND COURSES OF ACTION

OWN SITUATION

F. COURSES OF ACTION:

COA1- INSTALL REPEATER SYSTEM AT THE LOCATION OF GLOBE CELLSITE AT VIC GC 51P TT 937082 (CO-LOCATED)
COA2- INSTALL REPEATER SYSTEM AT SO TABAG-HAK AT VIC GC 51P TT 992107
COA3- TURN-OFF ALL C.E EQUIPMENT DURING THUNDERSTORMS AND LIGHTNING AND USE CELLULAR PHONES. LISTENING SILENCE EVERY 0600H, 1200H AND 0600H
COA3a- TURN-OFF ALL C.E EQUIPMENT DURING BAD WEATHER AND MESSENGER WILL BE OUR MEANS OF COMMUNICATIONS. LISTENING SILENCE EVERY 0600H, 1200H AND 0600H

PARA II - SITUATION AND COURSES OF ACTION

OWN SITUATION

F. COURSES OF ACTION:

COA4- DISRUPT THE ENEMY C3 USING LIMITED ELECTRONIC WARFARE.
COA5- DECEIVE ENEMY INFORMATION GATHERING THROUGH OBSERVANCE OF TOTAL RADIO LISTENING SILENCE
COA6-COORDINATION TO G6, 7ID, PA FOR THE ISSUANCE OF HARRIS TACTICAL RADIO (HF/VHF) AND SECURED MODE VHF REPEATER SYSTEM (SMVRS) TO FILL UP THE NEEDS OF OPERATING UNIT.
COA7- COORDINATION TO G6, 7ID, PA FOR ADDITIONAL COMMERCIAL RADIOS

PARA III - ANALYSIS OF OPPOSING COA

A. COMMUNICATION AND DISPOSITION OF OPPOSING FORCES:
B. SIGNAL PROPAGATION OF OPPOSING FORCES:
C. LIST OF ENEMY CAPABILITIES AND COUNTERMEASURE (RFI from Intel):
Capabilities:
Limitations:
D. LIST OF FRIENDLY FORCES CABABILITIES AND LIMITATION:
Capabilities:
Limitations:
E. COUNTER-MEASURES:
On the Enemy Capabilities:
Signal Platoon Limitations:

PARA III - ANALYSIS OF OPPOSING COA

OWN SITUATION

A. COMMUNICATION AND DISPOSITION OF OPPOSING FORCES

PARA III - ANALYSIS OF OPPOSING COA

OWN SITUATION

B. SIGNAL PROPAGATION OF OPPOSING FORCES:

PARA III - ANALYSIS OF OPPOSING COA

OWN SITUATION

C. LIST OF ENEMY CAPABILITIES AND COUNTERMEASURE (RFI FROM INTEL):

- Capabilities:

- Limitations:

PARA III - ANALYSIS OF OPPOSING COA

OWN SITUATION

D. LIST OF FRIENDLY FORCES CABABILITIES AND LIMITATION:

- CAPABILITIES:

- LIMITATIONS:

PARA III - ANALYSIS OF OPPOSING COA

OWN SITUATION

E. COUNTER-MEASURES:

- ON THE ENEMY Capabilities:
- Signal Platoon Limitations:

PARA IV - COMPARISON OF COAS

COA (TERRAIN)	ADVANTAGE	DISADVANTAGE
#1. INSTALL REPEATER SYSTEM AT THE LOCATION OF GLOBE CELLSITE AT VIC GC 51P TT 937082 (CO-LOCATED)	- CAN COVER SOME DEAD SPOTS IN THE AO	- NO CONTACTS ON LOW-LYING AREA -REQUIRES ADDITIONAL SECURITY
#2. INSTALL REPEATERS AT THE LOCATION OF GLOBE TOWER AND AT SO TABAG-HAK AT VIC GC 51P TT 992107	- CAN COVER DEAD SPOTS IN THE AO	-REQUIRES ADDITIONAL SECURITY

PARA IV - COMPARISON OF COAS

COA (WEATHER)	ADVANTAGE	DISADVANTAGE
#3. TURN-OFF ALL COMMUNICATION EQUIPMENT DURING BAD WEATHER AND USE **CELLULAR PHONES**. LISTENING SILENCE EVERY 0600H, 1200H AND 1800H	- SERVICEABILTY OF Communication EQUIPMENT ARE KEPT	- CELLULAR PHONES ELECTROMAGNETIC RADIATION ARE SUSCEPTIBLE TO LIGHTNING AND INTERCETION.

PARA IV - COMPARISON OF COAS

COA (WEATHER)	ADVANTAGE	DISADVANTAGE
#3a TURN-OFF ALL Communication EQUIPMENT DURING BAD WEATHER AND **MESSENGER WILL BE OUR PRIMARY MEANS OF COMMUNICATIONS**. LISTENING SILENCE EVERY 0600H, 1200H AND 1800H	- SERVICEABILTY OF Communication EQUIPMENT ARE KEPT	- SECURITY AND POSSIBLE COMPROMISE / CAPTURE OF MESSENGER

PARA IV - COMPARISON OF COAS

COA (ENEMY SITUATION)	ADVANTAGE	DISADVANTAGE
#4 DISRUPT THE ENEMY C3 USING LIMITED ELECTRONIC WARFARE.	-CONTROL THE ELECTROMAGNETIC SPECTRUM -IMPEDE ENEMY ASSAULTS VIA SPECTRUM	- LACK OF TRAINED PERSONNEL

PARA IV - COMPARISON OF COAS

COA (ENEMY SITUATION)	ADVANTAGE	DISADVANTAGE
#5 IMPLEMENTATION OF TOTAL RADIO SILENCE	-DECEIVED ENEMY INFORMATION GATHERING	NO C2 WILL BE ESTABLISHED

PARA IV - COMPARISON OF COAS

COA (OWN SITUATION)	ADVANTAGE	DISADVANTAGE
#6 COORDINATION TO _____, (UNIT) FOR THE ISSUANCE OF ADDITIONAL HARRIS TACTICAL RADIO (HF/VHF) AND SECURED MODE VHF REPEATER SYSTEM (SMVRS) TO FILL UP THE NEEDS OF OPERATING UNIT.	-FILL-UP THE LACK OF AVAILABLE CE EQUIPMENT	VERY LONG TIME FOR TRANSPORT WITH SECURITY IMPEDIMENTS.

PARA IV - COMPARISON OF COAS

COA (OWN SITUATION)	ADVANTAGE	DISADVANTAGE
#7 COORDINATION TO G6, 7ID, PA FOR ADDITIONAL COMMERCIAL RADIOS.	-FILL-UP THE LACK OF AVAILABLE COMMERCIAL RADIOS	-LIMITED RANGE -REQUIRES BASE RADIO AND/OR REPEATER -COMMS SECURITY COMPROMISE

PARA V - DECISION/RECOMMENDATIONS:

A. PERSONNEL REQUIREMENTS:
RECOMMENDATIONS:
B. C4S REQUIREMENTS:
RECOMMENDATIONS:
C. SUSTAINMENT REQUIREMENTS:
RECOMMENDATION:
D. MEANS OF COMMUNICATION:
E. LOCATION OF TCP AND MCP:
F. TASK LIST (Implied and Specific) (CMDR'S DECISION)

PARA V - DECISION/RECOMMENDATIONS:

A. PERSONNEL REQUIREMENTS:

RECOMMENDATIONS:

- (BE SPECIFIC) REQUIRED SKILLS NEEDED EX. CYBERSECURITY SPECIALIST

PARA V - DECISION/RECOMMENDATIONS:

B. C4S REQUIREMENTS:

RECOMMENDATIONS:

2 - HARRIS HH RADIO ETC. (SPECIFIC)

PARA V - DECISION/RECOMMENDATIONS:

C. SUSTAINMENT REQUIREMENTS:

RECOMMENDATION:

GENERATOR SET....
10 HARRIS HANDHELD....
ETC. (BE SPECIFIC)

PARA V - DECISION/RECOMMENDATIONS:

D. MEANS OF COMMUNICATION:

P
A
C
E

PARA V - DECISION/RECOMMENDATIONS:

E. LOCATION OF TCP AND MCP:

- 51PYK1234567890, Brgy._____Municipality/City/Province

PARA V - DECISION/RECOMMENDATIONS:

F. TASK LIST (Implied and Specific) (CMDR'S DECISION)

❑Install two (2) repeaters: SMVRS to SO Tan-awan and UHF repeater (Co-located) at Globe tower in order to maximize radio transmission in the area of operation. This will be installed by signal platoon.

❑During bad weather, turn-off all CE equipment and strictly observe radio silence every 0600h, 1200h and 1800h. This will prevent enemy in conducting limited electronic attack. This must be strictly implemented to the operating troops.

❑To counter enemy's limited capability of electronic warfare, coordinate with signal intelligence specialists for the conduct of limited electronic warfare before, during and after the conduct of operation. In the absence of electronic warfare specialist, strictly observe the given time for radio listening silence in order to prevent the enemy in detecting radio transmission of the operating troops.

❑Coordinate with G6, 7ID for issuance of additional twenty (20) VHF handheld radios, thirty (30) UHF radio, one (1) SMVRS and one (1) UHF base radio.